Assignments

Exercise 1: USART2 DMA Transfer

- Reproduce Demo 1 from the book:
  Configure USART2 with DMA1/CH7 and send a string to Realterm (or other terminal). The string should contain your student ID.
- Explain what the code
  ```
  huart.Instance->CR3 |= USART_CR3_DMAT;
  ```
  and
  ```
  huart.Instance->CR3 &= ~USART_CR3_DMAT;
  ```
  is "doing".
- Is there a HAL API that can be used instead?
- Add an LED toggle in the main loop to prove the CPU is free during the DMA transfer.

Exercise 2: Peripheral to RAM memory DMA Transfer

- Measure TMP36 once per second, store the ADC samples in a RAM buffer using DMA, convert to temperature, and display the values via USART2 using DMA (like in Ex1)
- Optional add-on: Store the ADC data in Flash memory, using DMA.

- Tips: Enable the correct DMA channels. Enable ADC1 with continuous conversion. Peripheral increment: disabled. Memory increment: enabled. Data width: Half-word (16 bits) for both Peripheral and Memory.
- Add an LED toggle in the main loop to prove the CPU is free during the DMA transfer.